An Offline Multimodal AI Orchestrator on a 2021 Foldable Phone

Project Tricorder: architecture, measured capabilities, and a comparison with contemporary on-device assistants

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Single-device case study. Not a vendor benchmark.

Abstract

This note describes a working, fully local multimodal stack on a Samsung Galaxy Z Fold 3 (Qualcomm Snapdragon 888). Four inference servers run concurrently in Termux and speak OpenAI-shaped HTTP: a 4B instruction-tuned language model for chat, a 500M vision-language model for camera understanding, a distilled Stable Diffusion checkpoint for text-to-image, and a Kokoro ONNX speech synthesizer. A thin Android WebView application owns the microphone, camera, sensors, and a single instrument UI. The contribution is not a new foundation model. It is an orchestration pattern that keeps chat, vision, image generation, and speech alive on one older flagship SoC without a cloud round-trip. On this device, inquiry and vision complete in seconds to tens of seconds; text-to-image at 384×384 and six sampler steps completes in just over two minutes; two-step sampling does not produce a finished picture. The design is portable to other aarch64 Termux phones as weights plus two CPU binaries plus an APK. Limitations are stated plainly: CPU-only diffusion, Termux operational burden, and no claim of uniqueness against OEM on-device assistants or other local-LLM apps.

1. What was built

Project Tricorder is a sideloaded Android application wrapping a single HTML/JS instrument panel. The panel does not contain the models. It routes user acts — typed or spoken inquiry, a camera loop labeled Sensor, a motion-triggered loop labeled Sentry, and an Imagine intent — to four loopback ports.

The servers, all process-local to Termux on 127.0.0.1, are:

Chat — llama-server, Gemma 3 4B Instruct Q4_K_M, port 8080, context 2048, three CPU threads.

Vision — llama-server, SmolVLM2-500M-Video-Instruct Q8_0 plus an F16 multimodal projector, port 8081, context 1024, two threads.

Image — sd-server from stable-diffusion.cpp, SD 2.1 Turbo Q4_0, port 8188, classifier-free guidance scale 1.

Speech — a small Python front end over Kokoro-82M ONNX (fp32) and voices-v1.0.bin, port 5000.

Speech input uses the phone's built-in Android SpeechRecognizer, not a second neural ASR we compiled.[1] Playback of Kokoro waveforms is native MediaPlayer in the APK, because WebView autoplay of fetched WAV was unreliable.

Hardware is a 2021 foldable: Snapdragon 888, Adreno 660, 12 GB RAM.[2] Diffusion and both llama.cpp instances run on CPU. A Vulkan or Adreno OpenCL rebuild of sd-server was considered and deferred: published Termux

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[1] G. Gerganov et al., llama.cpp, https://github.com/ggml-org/llama.cpp (accessed 30 Aug 2026). Android/Termux build notes: docs/android.md.

[2] leejet et al., stable-diffusion.cpp, https://github.com/leejet/stable-diffusion.cpp (accessed 30 Aug 2026). OpenAI-compatible sd-server: examples/server.

Vulkan attempts are often slower than CPU or fail to enumerate the GPU, and OpenCL needs an NDK sysroot dance that this study did not complete.

2. Why an orchestrator, not another chat app

Consumer on-device AI in 2026 is abundant and fragmented. PocketPal AI and Maid wrap llama.cpp for chat. MLC Chat is faster on some flagships and stricter about which models it will load.[3] Google ships Gemini Nano through AICore and a public Edge Gallery of Gemma-family models.[4] Samsung Galaxy AI and Apple Intelligence fold generation into the OS, with a cloud backstop and no user-owned GGUF slot. Image generation on phones is usually a separate APK (sd.cpp wrappers, “Local Diffusion”). Speech is another APK. Almost none of those products keep four heterogeneous backends hot and route them from one panel with shared hold/continuous semantics.

The engineering claim is narrower than “we invented on-device AI.” The claim is: concurrent, inspectable, OpenAI-compatible local services plus a single operator UI will run on a four-year-old Snapdragon if the operator accepts Termux as the runtime and minutes-scale diffusion.

3. Measured capabilities

3.1 Inquiry (language)

Spoken or typed questions go to Gemma 3 4B. The model is small enough for Q4_K_M on this SoC. Answers appear on a CRT-styled panel and are spoken by Kokoro after a sanitizer strips abbreviations and decimals that otherwise fragment phonemization (U.S.S. and 1013.2 were observed failure modes). A wake utterance beginning with “computer” is intercepted before the LLM and answered from device telemetry (uptime, battery, barometer when the sensor exists, a probe toward 8.8.8.8).

3.2 Sensor and Sentry (vision)

Sensor captures a still, sends it to SmolVLM2, prints and speaks a caption, then locks a Hold control so the loop cannot flood the TTS queue. Sentry diffs downscaled grayscale frames and only calls the VLM on a motion threshold. A separate “continuous machine vision (no Kokoro)” switch writes captions to the screen only. Those two controls were required; an early unthrottled loop saturated Kokoro and the UI.

3.3 Imagine (text-to-image)

sd-server’s OpenAI /v1/images/generations route honors size and ignores a top-level steps field. Step count must be embedded in the prompt as sd_cpp_extra_args.[5] On this checkpoint, 256×256 produced tiled or doubled subjects — expected when a 512-native UNet is asked for a latent that is too small. Two sampler steps produced color squares. Four steps at 512×512 produced a coherent image in about three minutes. Four steps at 384×384 looked good. Six steps at 384×384 looked better and finished in just over two minutes. Those timings are wall-clock on a warm Snapdragon 888, CPU only, one image, operator present. They are not vendor SLAs.

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[3] Google, Gemma 3 model card and ggml-org/gemma-3-4b-it-GGUF on Hugging Face. Gemma Terms of Use apply.

[4] Hugging Face TB, SmolVLM2-500M-Video-Instruct; GGUF pack ggml-org/SmolVLM2-500M-Video-Instruct-GGUF (Q8_0 + mmproj-f16).

[5] Stability AI, SD-Turbo (Adversarial Diffusion Distillation). GGUF used: gpustack/stable-diffusion-v2-1-turbo-GGUF, Q4_0.

3.4 Operations

A Termux script starts sshd on port 8022 and the four servers under setsid/nohup plus a wake-lock. Starting them from an SSH TTY and then dropping the session was observed to reap the process group. Starting them from the Termux app on the device was stable enough for a working day. Health lamps in the APK poll /health or GET / every four seconds.[6]

4. Value

Privacy. Camera frames, prompts, and spectrograms never leave the handset unless the operator later chooses to. That is the load-bearing property for field notes, clinical-adjacent dictation, legal holds, and any site where a radio is forbidden or untrusted.

Continuity on old silicon. Gemini Nano 4's published device list in late August 2026 is seven phones, none of them a 2021 Fold.

[7] A GGUF sidecar does not wait for an OEM allow-list.

Inspectability. Four ports can be curled from a desk. Failures show up as a red lamp and a log line, not as a silent cloud 500.

Pedagogy. The stack is made of named open engines and named weights. A student can replace Gemma with another GGUF without rewriting the APK.

What it is not. It is not a replacement for a datacenter 70B model. It is not real-time image generation. It is not an unattended appliance; Termux, battery exemption, and a first-start from the phone are part of the product.

5. Use cases that fit the measurements

Field instrument. Point the cover camera, Hold, hear a scene description, ask a follow-up. Works in a Faraday tent or a basement.

Private briefing. Dictate a question, get a short spoken answer, no tenant log.

On-site sketch. "Imagine a red tricorder on a desk" at 384×6 when two minutes is acceptable and the radio is not.

Teaching edge inference. One device demonstrates quantization, mmproj wiring, turbo distillation step counts, and why 256 tiles on an SD 2.1 backbone.

Accessibility experiment. Local TTS plus local STT with no account. Quality is OEM-ASR plus Kokoro, not a studio voice.

Use cases that do not fit: live translation of a meeting, photoreal 1024^2 on demand, medical diagnosis, anything that needs a model larger than this RAM budget.

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[6] hexgrad, Kokoro-82M; ONNX runtime via thewh1teagle/kokoro-onnx model-files-v1.0 (fp32 graph + voices-v1.0.bin). PocketPal AI ships a related Kokoro pack.

[7] Android SpeechRecognizer / RecognizerIntent; on-device recognition where the OEM pack is installed. Not a learned ASR we trained.

6. Limitations

Single device, single operator, no blinded comparison against PocketPal or Edge Gallery on the same prompt set. Image times will move with thermal state. Gemma 3 weights are license-gated. Termux binaries are aarch64-Termux portable, not Play-Store portable. The APK is a debug build. Sensor fusion beyond barometer/battery/uptime was not the focus. GPU offload remains unproven here.

Table 1. Contemporary phone-side AI systems versus this orchestrator (features as publicly described, August 2026).

7. Comparison matrix

Rows mix products, OEM stacks, and toolchains. "Yes" means the capability exists in the common configuration, not that quality is equal. OGAM's published description is the nearest all-in-one product claim; it was not timed on the Fold 3.[8]

System	Local chat	Vision VLM	Text-to-image	Speech I/O	Local HTTP APIs	BYO weights	Notes
Project Tricorder (this work)	Yes — Gemma 3 4B Q4_K_M via llama-server	Yes — SmolVLM2-500M + mmproj	Yes — SD 2.1 Turbo Q4 via sd-server	STT: Android SpeechRecognizer. TTS: Kokoro-82M ONNX	Yes — :8080 :8081 :8188 :5000 concurrent	Yes — GGUF / ONNX files	WebView APK + Termux sidecar. Measured on Snapdragon 888.
PocketPal AI	Yes — llama.cpp GGUF	Varies by model pack	No (chat/TTS focus)	Yes — ships Kokoro-class TTS	App-internal; not a multi-port orchestrator	Yes — any GGUF	Strong consumer local-chat app. CPU/Vulkan. No NPU.
MLC Chat	Yes — TVM-compiled catalog	Limited to compiled set	No	Limited	No general OpenAI-compat sidecar	No — precompiled list only	Fast on flagship Adreno/Hexagon. Less model freedom.
Google AI Edge Gallery / Gemini Nano	Yes — Gemma / Nano family	Yes on supported SoCs (multimodal Nano/Gemma 4)	No general local SD	On-device ASR in ML Kit GenAI	AICore / ML Kit APIs, not user-owned ports	No — Google catalog	Best integration on Pixel / listed Samsungs. Hardware-gated (Nano 4: seven phones as of Aug 2026).
Samsung Galaxy AI	Cloud + on-device mix	On-device features vary by year	Cloud drawing tools, not local SD.cpp	On-device + cloud	No user HTTP stack	No	OEM assistant. Not a bring-your-own-weights orchestrator.
Apple Intelligence	On-device + Private Cloud Compute	On-device visual intelligence	No local SD.cpp	Yes	No user HTTP stack	No	Closed stack. Not Android-portable.
OGAM / Off Grid AI	Yes — GGUF	Yes — claimed	Yes — claimed on-device SD	Whisper + speech claimed	Local-network servers claimed	Yes — GGUF	Closest product-shaped analogue. Broader claimed surface; not the measured stack here.
Maid / ChatterUI / Layla	Yes — llama.cpp wrappers	Rare / model-dependent	No	App TTS or none	Usually none	Often yes (GGUF)	Chat clients. Not multimodal orchestrators.
Termux + Ollama / llama.cpp	Yes	If mmproj wired	Only if sd.cpp also built	Only if separately built	Yes if you bind ports	Yes	The substrate this work sits on. No unified instrument UI by default.
sdcpp-on-android / Local Diffusion	No	No	Yes — sd.cpp	No	Optional	Yes — SD weights	Image-only clients.

Table 1. Feature presence, not quality scores. Sources: vendor docs and the reviews cited in notes 9–13.

[8] Qualcomm Snapdragon 888 / Adreno 660, Samsung Galaxy Z Fold 3. CPU-only ggml backends unless OpenCL/Vulkan is rebuilt.

8. Closing

A 2021 foldable can host four local generative services at once if the operator treats the phone as a small computer: Termux, GGUF files, and an honest budget for diffusion time. The interesting layer is the orchestrator — hold semantics, concurrent ports, a shared speech channel, and sliders that map onto the actual sd-server protocol — not any one weight file.

LinkedIn takeaway, still factual: this is a case study in composition. The models already existed. The phones already existed. Wiring them so a person can ask, look, imagine, and hear an answer with the radios off is the part that was missing on this device, and that part now runs.

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